

Engineering Physics (2025)
Course code 25PY101
Module 2 Unit 1: Quantum theories of solids

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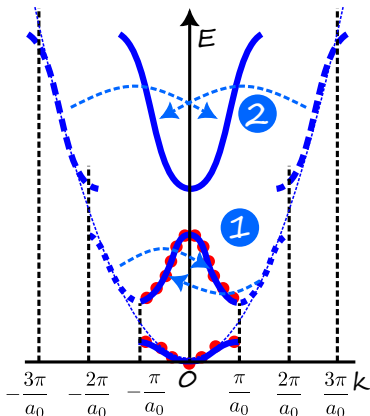
Unit 2 Plan

- 1 Quantum Free Electron Theory
- 2 Fermi-Dirac distribution
- 3 Electronic specific heat of solids
- 4 Density of states (qualitative)
- 5 Success and Failures of quantum free electron theory of solids
- 6 E-k diagram
- 7 Classification of materials based on bands in solids
- 8 Fermi level in semiconductors- intrinsic and extrinsic

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Reduced band dispersion – Band structure

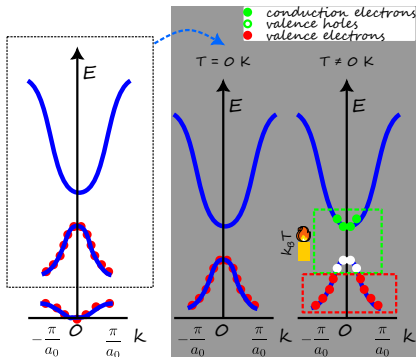


- Due to the periodicity of lattice, the band dispersion can be analyzed within the **zone** from $k = -\frac{\pi}{a_0}$ to $k = +\frac{\pi}{a_0}$.
- The original band dispersion can be reduced to the zone by folding the “arms” of the parabola resulting in the reduced band dispersion. This process is called **zone folding**.
- The reduced band dispersion is called the **band structure** of the material and it is the signature of the material.

Key Insight

Bandstructure is the fingerprint or “DNA” of a crystalline solid.

Band structure – Concept of hole



- For a semiconductor, due to the presence of energy band gap, there are no free electrons at $T = 0$ K.
- At $T \neq 0$ K, the electrons near the valence band maximum thermalize and jump to the levels near the conduction band minimum. This results in quantum vacancy in the valence band. This is called **hole**.
- Conduction electron and valence hole contribute to conduction.

Key Insight

Hole is the absence of electron in the valence band.

Parabolic approximation – Effective mass

- For the conduction band, the dispersion relation can be approximated with an upward facing parabola given by

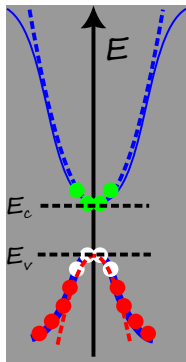
$$E = E_c + \frac{\hbar^2 k^2}{2m_n^*}$$

where m_n^* is the **effective electron mass** and E_c is the minimum of conduction band.

- Similarly, for the valence band, the dispersion relation can be approximated with a downward facing parabola given by

$$E = E_v - \frac{\hbar^2 k^2}{2m_p^*}$$

where m_p^* is the **effective hole mass** and E_v is the maximum of valence band.



Density of states in conduction band

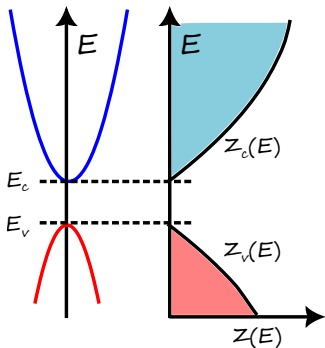
- For the conduction band, dispersion relation can be written as

$$E - E_c = \frac{\hbar^2 k^2}{2m_n^*}$$

so that the form is similar to the free electron dispersion relation. Thus, the density of states in the conduction band $Z_c(E)$ is given by

$$Z_c(E) = \alpha^* \sqrt{E - E_c}$$

where α^* is a constant with expression that contains universal constants and m_n^* ^a.



^aFrom QFET slides, we can infer $\alpha^* = 4\pi \left(\frac{2m_n^*}{\hbar^2} \right)^{3/2}$

Density of states in valence band

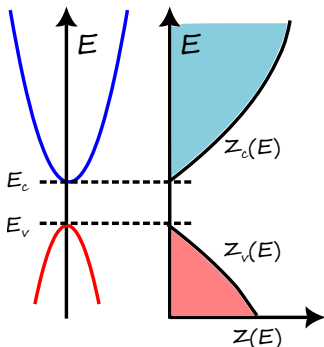
- Similarly for the valence band, dispersion relation can be written as

$$E_v - E = \frac{\hbar^2 k^2}{2m_p^*}$$

so that the density of states in the valence band $Z_v(E)$ is given by

$$Z_v(E) = \beta^* \sqrt{E_v - E}$$

where β^* is a constant with expression that contains universal constants and m_p^* ^a.



^aFrom QFET slides, we can infer $\beta^* = 4\pi \left(\frac{2m_p^*}{\hbar^2} \right)^{3/2}$.

Electron carrier concentration

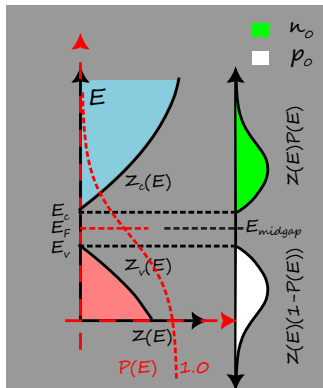
- The conduction electron density n_0 is given by

$$\begin{aligned} n_0 &= \int_{E_c}^{\infty} Z_c(E) P(E) dE \\ &= \int_{E_c}^{\infty} \frac{\alpha^* \sqrt{E - E_c}}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} dE \end{aligned}$$

- By applying Boltzmann approximation and the exact expression for α^* , it can be shown that

$$n_0 = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)$$

where N_c is the effective density of states in the conduction band.



Hole carrier concentration

- Similarly the valence hole density p_0 is given by

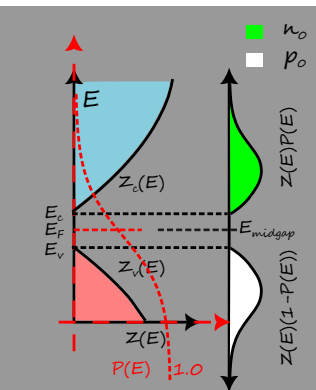
$$p_0 = \int_{E_c}^{\infty} Z_c(E) (1 - P(E)) dE$$

$$= \int_0^{E_v} \beta^* \sqrt{E_v - E} \left(1 - \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} \right) dE$$

- By applying **Boltzmann approximation** and the exact expression for α^* , it can be shown that

$$p_0 = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

where N_v is the effective density of states in the valence band.



Intrinsic carrier concentration – $n_0 p_0$ product

- For an intrinsic semiconductor, electron-hole pairs are generated at $T \neq 0$ K. So the conduction electron density is equal to valence hole density

$$n_0 = p_0 = n_i$$

where n_i is defined as the intrinsic carrier concentration.

- The intrinsic carrier concentration is given by the product of electron and hole concentrations

$$\begin{aligned} n_0 p_0 = n_i^2 &= N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \cdot N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) \\ &= N_c N_v \exp\left(-\frac{E_c - E_F + E_F - E_v}{k_B T}\right) \\ n_i^2 &= N_c N_v \exp\left(-\frac{E_c - E_v}{k_B T}\right) \end{aligned}$$

Intrinsic carrier concentration

- Therefore, the $n_0 p_0$ is independent of Fermi energy level and is given in terms of the material properties and temperature.

$$n_0 p_0 = n_i^2$$

where n_i is given by

$$\begin{aligned} n_i &= \sqrt{N_c N_v \exp\left(-\frac{E_c - E_v}{k_B T}\right)} \\ &= \sqrt{N_c N_v} \exp\left(-\frac{E_c - E_v}{2k_B T}\right) \\ &= \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right) \end{aligned}$$

where E_g is the energy band gap and is given by

$$E_g = E_c - E_v$$

$$\therefore n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)$$

Fermi level of intrinsic semiconductor

- Since the electron and hole exist in pairs in an intrinsic semiconductor

$$n_0 = p_0$$

$$\Rightarrow N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

- Taking logarithm on both sides of the equation, we have

$$\ln N_c + \ln \left[\exp\left(-\frac{E_c - E_F}{k_B T}\right) \right] = \ln N_v + \ln \left[\exp\left(-\frac{E_F - E_v}{k_B T}\right) \right]$$

$$\Rightarrow \ln N_c - \frac{E_c - E_F}{k_B T} = \ln N_v - \frac{E_F - E_v}{k_B T}$$

$$\Rightarrow \frac{2E_F}{k_B T} = \frac{E_c + E_v}{k_B T} + \ln N_v - \ln N_c$$

$$\Rightarrow E_F = \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \frac{N_v}{N_c}$$

- If we substitute the expressions for N_c and N_v , in the ratio of N_c over N_v all the terms cancel out except the effective masses of electron and hole.

Fermi level of intrinsic semiconductor

- The expressions for effective density of states are

$$N_c = 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2}, \quad N_v = 2 \left(\frac{2\pi m_p^* k_B T}{h^2} \right)^{3/2}$$

- Therefore, the Fermi level in terms of effective masses of charge carriers is given by

$$\begin{aligned} E_F &= \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \left(\frac{m_p^*}{m_n^*} \right)^{3/2} \\ &= E_{\text{midgap}} + \frac{3k_B T}{4} \ln \frac{m_p^*}{m_n^*} \end{aligned}$$

where E_{midgap} is defined as the midgap energy level that is in the midpoint of conduction band minimum and valence band maximum and is given by

$$E_{\text{midgap}} := \frac{E_c + E_v}{2}$$

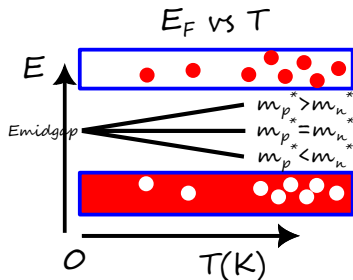
$$\therefore E_F = E_{\text{midgap}} + \frac{3k_B T}{4} \ln \frac{m_p^*}{m_n^*}$$

E_F vs T of intrinsic semiconductor

- Since the Fermi energy level varies with temperature, let us consider two cases – $T = 0\text{ K}$ and $T \neq 0\text{ K}$.
- Also, the Fermi energy level varies with effective masses of electron and hole. So let us consider three cases – $m_p = m_n$, $m_p > m_n$, and $m_p < m_n$.

	$T = 0\text{ K}$	$T \neq 0\text{ K}$
$m_p = m_n$	E_{midgap}	E_{midgap}
$m_p > m_n$	E_{midgap}	$E_{\text{midgap}} + \delta$
$m_p < m_n$	E_{midgap}	$E_{\text{midgap}} - \delta$

Table: Here δ is a positive quantity that linearly increases with temperature.



Key Insight

With temperature, E_F deviates from E_{midgap} for semiconductors with differing electron and hole effective masses.

Conductivity of intrinsic semiconductor

- Conduction in a semiconductor is through two channels – electron and hole. Therefore

$$\sigma_i = \sigma_e + \sigma_h$$

- For each channel, the conductivity is given by the product of charge carrier concentration and mobility. Therefore

$$\begin{aligned}\sigma_i &= n_0 e \mu_e + p_0 e \mu_h \\ &= n_i e (\mu_e + \mu_h)\end{aligned}$$

$$\therefore \sigma_i = n_i e (\mu_e + \mu_h)$$

- If we express the intrinsic carrier concentration in terms of material properties, then

$$\begin{aligned}\sigma_i &= \sqrt{N_c N_v} e (\mu_e + \mu_h) \exp\left(-\frac{E_g}{2k_B T}\right) \\ &= \sigma_0 \exp\left(-\frac{E_g}{2k_B T}\right)\end{aligned}$$

where σ_0 is a constant in terms of material properties.

Estimation of band gap of intrinsic semiconductor

- Therefore, the variation of conductivity of intrinsic semiconductor with temperature is given by

$$\therefore \sigma_i = \sigma_0 \exp\left(-\frac{E_g}{2k_B T}\right)$$

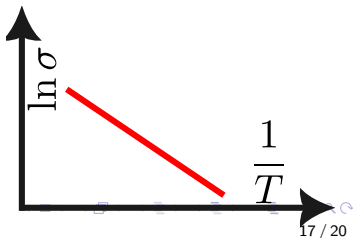
- If we take the logarithm on both sides of above equation, then

$$\begin{aligned}\ln \sigma_i &= \ln \sigma_0 + \ln \left[\exp\left(-\frac{E_g}{2k_B T}\right) \right] \\ &= \ln \sigma_0 - \frac{E_g}{2k_B T}\end{aligned}$$

- Therefore, a plot of $\ln \sigma$ vs $\frac{1}{T}$ will be straight line with slope m given by

$$m = -\frac{E_g}{2k_B}$$

From the slope of the plot, we can estimate the energy band gap.



Some useful information for calculations

	Si	GaAs	Ge
μ_n (cm ² V ⁻¹ s ⁻¹)	1350	8500	3900
μ_p (cm ² V ⁻¹ s ⁻¹)	480	400	1900

Table: Mobility of carriers in Si, GaAs, and Ge at 300 K. Taken from Section 5.1 of Neamen

	Si	GaAs	Ge
n_i (cm ⁻³)	1.5×10^{10}	1.8×10^6	2.4×10^{13}

Table: Intrinsic carrier concentration in Si, GaAs, and Ge at 300 K.

Problems on Fermi level of intrinsic semiconductor

Problem

A quantum state in the conduction band is $k_B T$ above conduction band minimum E_c of an intrinsic semiconductor. The Fermi level is 0.25 eV below E_c . Find the probability that the quantum state is occupied at 300 K.

Problem

For the same material in above problem, if the effective density of states in conduction band N_c is $2.8 \times 10^{19} \text{ cm}^{-3}$ at 300 K, then find the electron concentration n_0 .

Problem

Calculate the position of the intrinsic Fermi level with respect to the center of the bandgap in Si at 300 K.

Problem

Find the resistivity of intrinsic Ge at 300 K. [Hint: Check useful info slide]

Problems on Fermi level of intrinsic semiconductor

Problem

The resistivity of intrinsic Si is $2.3 \times 10^3 \Omega \text{ m}$ at 300 K. Calculate its resistivity at 100 °C. [Hint: $E_g^{\text{Si}} = 1.12 \text{ eV}$]

Tough problems

Problem

Calculate the thermal equilibrium electron concentration in Si at 400 K. N_c at 300 K is $2.8 \times 10^{19} \text{ cm}^{-3}$. [Hint: Solve the below problem.]

Problem

Calculate the ratio of conduction electrons at 400 K to conduction electrons at 300 K. [Hint: From the explicit expression for N_c , observe the form of N_c vs T .]